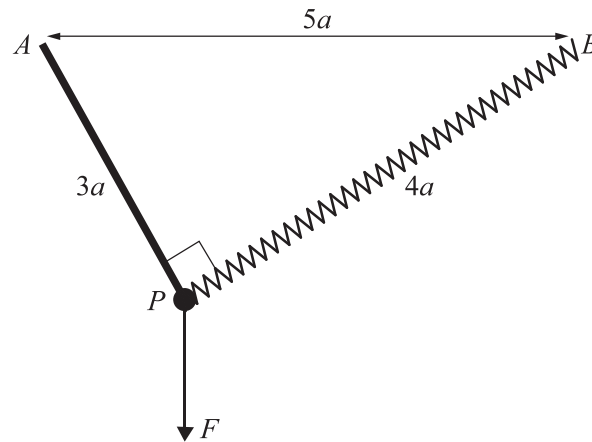




A particle P of mass m is attached to one end of a light rod of length $3a$. The other end of the rod is able to pivot smoothly about the fixed point A . The particle is also attached to one end of a light spring of natural length a and modulus of elasticity kmg . The other end of the spring is attached to a fixed point B . The points A and B are in a horizontal line, a distance $5a$ apart, and these two points and the rod are in a vertical plane.

Initially, P is held in equilibrium by a vertical force F with the stretched length of the spring equal to $4a$ (see diagram). The particle is released from rest in this position and has a speed of $\frac{6}{5}\sqrt{2ag}$ when the rod becomes horizontal.

- (a) Find the value of k . [5]

[illegible]

- (b) Find F in terms of m and g . [2]

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- (c) Find, in terms of m and g , the tension in the rod immediately before it is released. [2]

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